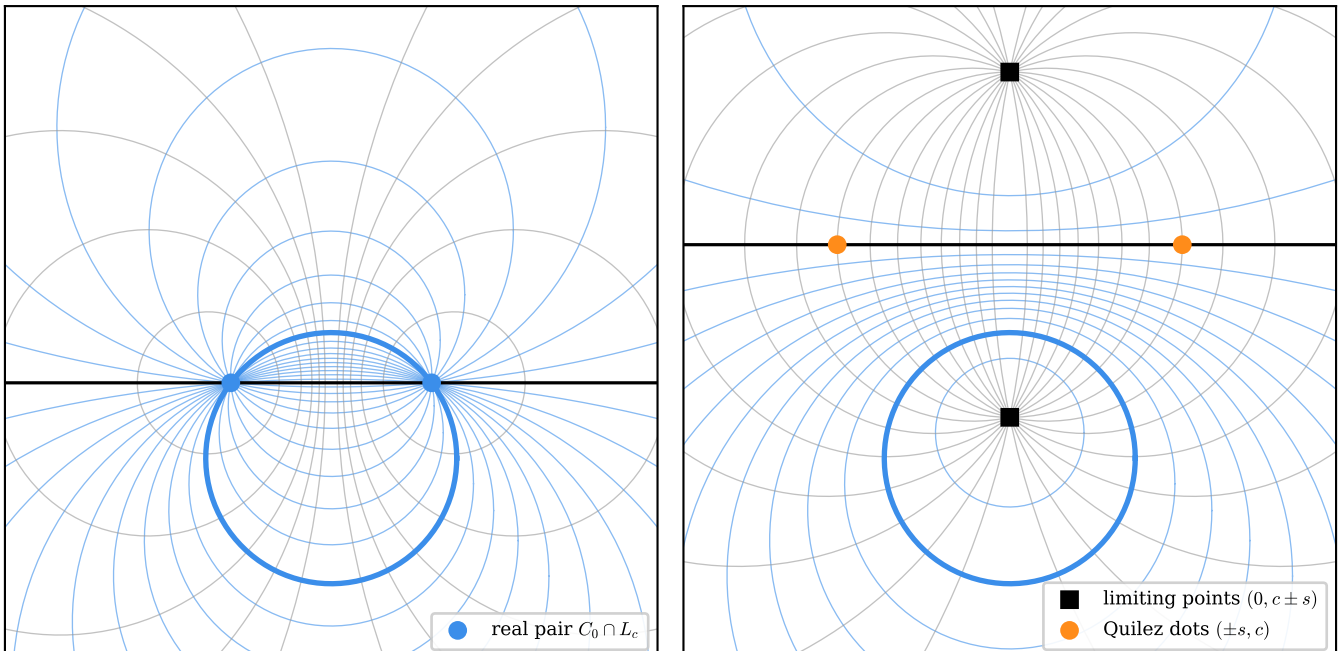

$c = 0.6 < r$: elliptic pencil, real pair;
limiting points $(0, c \pm i\,0.8)$ imaginary
real pair $C_0 \cap L_c$
$c = 1.7 > r$: hyperbolic pencil, pair $(\pm is, c)$, $s = 1.37$;
limiting points real, inverse in C_0
limiting points $(0, c \pm s)$
Quilez dots $(\pm s, c)$